

# AIGR Deep Research — id3-a (Q91399, *Xenopus laevis*)

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**Focus: proposed new GO term "neural crest progenitor maintenance"**

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Hypothesis slug: nc-progenitor-maintenance-scrna

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## Executive Judgment

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**Verdict: PARTIALLY SUPPORTED (biology real; proposed evidence type weak; proposed bespoke GO term not recommended as framed).**

Two separable claims are bundled in the seed:

1. **Functional claim** — *id3 maintains the neural-crest (NC) progenitor pool*. This is **well supported** by direct loss- and gain-of-function experiments in *Xenopus* (Kee & Bronner-Fraser 2005,

**P** 15769946

Id3 knockdown eliminates NC precursors through cell-cycle arrest and apoptosis of the progenitor pool "rather than a cell fate switch," and Id3 overexpression expands the NC domain. This is the strongest, most direct evidence and it is organism-appropriate.

2. **Model claim + proposed test** — *the competence factors id3, myc, hes4 (hairy2), pou5f3 and lin28 are expressed continuously from blastula animal-cap cells into premigratory NC (progenitors "maintain" rather than "switch off and re-acquire"), testable by scrNA expression-continuity in the Briggs et al. 2018 time course* (

**P** 29700227

This is **unresolved and the chosen test is under-powered for the curation question**. Expression continuity is correlative and cannot by itself establish a *maintenance function*; the retain-vs-reacquire question is explicitly framed as open by the same research

community (Schock, York & LaBonne 2023 review,  
 P 35534333  
 and cross-species data show the premise is not universal (lamprey *pou5* is **absent** from NC;  
 York et al. 2024,  
 P 39060477

**Most important caveats for the curator:** - The current annotation state already covers the **biology**. A programmatic pull of UniProt Q91399 (ID3A\_XENLA) shows id3-a already carries **GO:0014029 neural crest formation (IMP)** and **GO:0051726 regulation of cell cycle (IMP)** — exactly the proliferation/pool-maintenance role from Kee & Bronner-Fraser 2005 — plus a precise MF set (**GO:0043425 bHLH transcription factor binding (IPI)**, GO:0046983 protein dimerization activity, GO:0003714 transcription corepressor activity). A new "neural crest progenitor maintenance" term would therefore be **redundant**. - id3's *molecular function* is a **general HLH inhibitor of bHLH/E-proteins** (sequestration that blocks their DNA binding), active across many stem/progenitor contexts (hESC, spermatogonia, erythroid, osteoblast). "Neural crest progenitor maintenance" is a **context-specific developmental outcome (BP)**, not the gene product's primary molecular activity. - A brand-new NC-specific GO term risks over-specificity and conflating a downstream developmental phenotype with the core function. - I did **not** re-run the Briggs scRNA analysis. I verified the dataset is public (**GEO GSE113074**, 136,966 *X. tropicalis* single cells over 10 timepoints, including neural-plate-enriched libraries) but a full reanalysis (download of large count matrices, *X. tropicalis* ortholog mapping, trajectory re-derivation) was out of scope and I will not fabricate a result. The judgment rests on primary functional literature, direct UniProt/GO database evidence, and published synthesis — including Briggs 2018, which itself "assess[es] conflicting models of neural crest development."

## Evidence Matrix

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| # | Citation                                                   | Evidence type                           | Stance                       | Claim tested                                | Key finding                                                                                                                                          | Context                                                |
|---|------------------------------------------------------------|-----------------------------------------|------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 1 | Kee & Bronner-Fraser 2005,<br><a href="#">P 15769946</a>   | Mutant (MO) + overexpression phenotype  | <b>Supports</b> (function)   | id3 maintains NC progenitor pool            | Id3 MO → absence of NC precursors via cell-cycle inhibition + death of progenitor pool, <i>not</i> a fate switch; Id3 overexpression expands NC      | <i>Xenopus</i> , neural plate border, gastrula/neurula |
| 2 | Nichane et al. 2008,<br><a href="#">P 18710660</a>         | Mutant/structure-function               | Supports (co-factor)         | HLH factors maintain pre-NC state           | Hairy2 (hes4) "essential for neural crest progenitor survival and maintains cells in a mitotic undifferentiated pre-neural crest state"              | <i>Xenopus</i> , NP border                             |
| 3 | York et al. 2024,<br><a href="#">P 39060477</a>            | Comparative / evolutionary + functional | <b>Qualifies / competing</b> | Continuous blastula→NC program is conserved | Shared blastula–NC GRN is ancient, BUT lamprey <i>pou5</i> is "absent from neural crest" — continuity of individual factors is <b>not universal</b>  | <i>Xenopus</i> vs lamprey                              |
| 4 | Schock, York & LaBonne 2023,<br><a href="#">P 35534333</a> | Review (labeled synthesis)              | <b>Qualifies</b>             | Retain vs re-acquire                        | States pluripotency is " <b>either</b> retained in the neural crest from blastula stages <b>or</b> subsequently reactivated" — explicitly unresolved | Vertebrate NC                                          |
| 5 |                                                            |                                         | Qualifies (data source)      | Continuity along                            | Whole-embryo time course; "assess conflicting models of                                                                                              | <i>X. tropicalis</i>                                   |

| # | Citation                                                 | Evidence type                          | Stance                                | Claim tested                             | Key finding                                                                                                                             | Context          |
|---|----------------------------------------------------------|----------------------------------------|---------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------|
|   | Briggs et al. 2018,<br><a href="#">P 29700227</a>        | Primary scRNA atlas (proposed dataset) |                                       | inferred trajectory                      | neural crest development" — relevant but continuity is correlative                                                                      | whole embryo     |
| 6 | Jiang et al. 2022,<br><a href="#">P 35701409</a>         | Direct assay (KO, scRNA)               | Supports mechanism / shows generality | id3 maintenance activity is NC-specific? | ID1/ID3 KO decreases survival and pluripotency of hESCs via TCF3→AKT; function is a <b>general</b> stem-cell survival/pluripotency role | Human ESC        |
| 7 | Li, Du & Jin 2024,<br><a href="#">P 39342788</a>         | Molecular/ structural                  | Supports (MF)                         | id3 molecular activity                   | ID1/ID3 "negatively regulate bHLH transcription factors by forming heterodimers"; roles in neurodevelopment, cardiovascular, metastasis | Human ESC lines  |
| 8 | Liu et al. 2026,<br><a href="#">P 42733085</a>           | Molecular (ChIP, functional)           | Supports (MF)                         | id3 molecular activity                   | Id3 "binds to bHLH transcription factors (e.g., E-proteins) to suppress their DNA-binding capacity"; acts as a brake on differentiation | BMSC/ osteoblast |
| 9 | Huber, Rao & LaBonne 2024,<br><a href="#">P 38884356</a> | Perturbation (epigenetic)              | Supports (model)                      | NC retains broad potential               | BET inhibition → loss of pluripotency at blastula and loss of NC at neurula; links blastula potency to NC                               | <i>Xenopus</i>   |

| #  | Citation                                                                  | Evidence type                       | Stance                        | Claim tested                        | Key finding                                                                                                                                                                                                             | Context                              |
|----|---------------------------------------------------------------------------|-------------------------------------|-------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 10 | Rigney, York & LaBonne 2025, <a href="#">P 40292574</a> / <b>39868152</b> | Perturbation                        | Supports (model)              | Shared blastula–NC GRN              | Klf2/Klf17 regulate exit from pluripotency and NC boundary; NC "shares significant gene regulatory architecture with pluripotent blastula stem cells"                                                                   | <i>Xenopus</i> , lamprey             |
| 11 | <b>UniProt Q91399 (ID3A_XENLA)</b> — current GO annotations               | Database record                     | <b>Qualifies (redundancy)</b> | Is the proposed term novel/ needed? | id3-a <b>already</b> annotated: BP GO:0014029 neural crest formation (IMP), GO:0051726 regulation of cell cycle (IMP); MF GO:0043425 bHLH TF binding (IPI), GO:0046983 dimerization, GO:0003714 corepressor; CC nucleus | <i>X. laevis</i> gene product record |
| 12 | <b>GEO GSE113074</b> (Briggs 2018 series record)                          | Dataset metadata                    | Qualifies (feasibility)       | Can the proposed scRNA test be run? | Public; 136,966 <i>X. tropicalis</i> cells, 10 timepoints, incl. 4 neural-plate-enriched libraries; summary states it "assess[es] conflicting models of neural crest development"                                       | <i>X. tropicalis</i> whole embryo    |
| 13 | <b>UniProt Q91399 sequence</b>                                            | Structural/ evolutionary (sequence) | <b>Qualifies (MF scope)</b>   | Does Id3 bind DNA or                | Id3-a lacks the bHLH basic region (max 13-aa K/R fraction 0.15)                                                                                                                                                         | <i>X. laevis</i> / cross-            |

| # | Citation        | Evidence type | Stance | Claim tested    | Key finding                                                            | Context          |
|---|-----------------|---------------|--------|-----------------|------------------------------------------------------------------------|------------------|
|   | (computed here) |               |        | sequester bHLH? | vs canonical DNA-binding MYOD1 (0.54, "DRRKAATMRERRR"); human ID3 0.31 | species sequence |

## GO Curation Implications (leads — require curator verification)

**Recommendation:** do NOT mint a bespoke "neural crest progenitor maintenance" term for **id3-a**. The decisive point is that the supportable biology is *already annotated* on Q91399. Full decision table is saved as the artifact `id3a_GO_decision_table.csv`. Summary:

| Aspect | GO term                                            | Current status on Q91399   | Suggested action                                                                                                                                       |
|--------|----------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| MF     | GO:0043425 bHLH transcription factor binding       | <b>Present (IPI)</b>       | <b>Retain</b> — core molecular activity (E-protein/ bHLH sequestration)                                                                                |
| MF     | GO:0046983 protein dimerization activity           | Present (IEA)              | Retain — HLH-mediated heterodimerization mechanism                                                                                                     |
| MF     | GO:0003714 transcription corepressor activity      | Present (IBA)              | Retain                                                                                                                                                 |
| BP     | GO:0043392 negative regulation of DNA binding      | <b>Present (IDA)</b>       | Retain — direct mechanism                                                                                                                              |
| BP     | GO:0014029 <b>neural crest formation</b>           | <b>Present (IMP)</b>       | <b>Retain</b> — this already covers the seed's NC role (from <a href="#">P 15769946</a> )                                                              |
| BP     | GO:0051726 <b>regulation of cell cycle</b>         | <b>Present (IMP)</b>       | <b>Retain</b> — this already covers the "maintenance/proliferation" biology                                                                            |
| BP     | GO:0019827 stem cell population maintenance        | Absent                     | <i>Optional add</i> only if curator wants explicit "maintenance" wording — annotate with a neural-crest extension, IMP from <a href="#">P 15769946</a> |
| BP     | GO:0032922 circadian regulation of gene expression | Present (IEA, TreeGrafter) | Flag as non-core / pipeline transfer                                                                                                                   |
| CC     | GO:0005634 nucleus                                 | Present (IBA)              | Retain                                                                                                                                                 |

**On the proposed new term specifically:** "neural crest progenitor maintenance" is (a) **redundant** — GO:0014029 (neural crest formation, IMP) + GO:0051726 (regulation of cell cycle, IMP) already encode the exact Kee & Bronner-Fraser 2005 phenotype; (b) **too narrow/organism-flavored** relative to id3's pleiotropic bHLH-antagonist MF; and (c) **not established**

by the proposed evidence type (scRNA co-expression is correlative). If curators still want an explicit "maintenance" concept, prefer adding existing **GO:0019827 stem cell population maintenance** with an "occurs in / part\_of neural crest" annotation extension (IMP,

P 15769946

rather than creating a standalone ontology term. The pluripotency-retention ("maintain vs re-acquire") framing belongs at **review/GRN level**, not as a gene-specific GO term, because it is unresolved and not established by direct assay for id3.

## Mechanistic Scope

- **Immediate molecular function (direct):** id3 is an Inhibitor-of-DNA-binding HLH protein. It heterodimerizes with class-I bHLH E-proteins (e.g., TCF3/E2A) through its HLH domain and prevents them from binding E-box DNA — i.e., it **sequesters/inactivates bHLH transcription factors**

( P 39342788

P 42733085

It does not itself bind DNA. **This was confirmed here at the sequence level:** Q91399 lacks the arginine/lysine-rich basic region that canonical DNA-binding bHLH proteins carry adjacent to the HLH (max 13-aa K/R fraction 0.15 for Id3-a vs 0.54 for MYOD1, whose basic region is "DRRKAATMRERRR"). Hence any sequence-specific **DNA-binding MF would be incorrect**; the supported MF is GO:0043425 bHLH transcription factor binding / GO:0046983 protein dimerization activity.

- **Immediate cellular consequence (direct, Xenopus):** keeps NC progenitors cycling and alive; its loss triggers cell-cycle arrest and apoptosis of the progenitor pool

( P 15769946

- **Downstream / developmental outcome (inferred, not core function):** maintenance of the NC progenitor domain, broad multi-germ-layer potential, and ultimately NC derivatives. "Neural crest progenitor maintenance" belongs here — it is a **BP-level outcome of the molecular activity in a specific tissue**, not the molecular activity itself.
- **Model-level claim (unresolved):** continuous competence-factor expression from blastula into NC as evidence that NC *retains* (vs re-acquires) a pluripotency program.

## Conflicts and Alternatives

1. **Correlation ≠ function.** The seed's proposed scRNA continuity/co-expression test, even if positive, would show *co-expression*, not *maintenance function*. The maintenance function is already demonstrated more convincingly by perturbation  
(P 15769946).
2. **Species variability undercuts universality.** Lamprey *pou5* is absent from NC  
(P 39060477),  
so "all five factors continuously expressed blastula→NC" is not a vertebrate-general fact; it is at best lineage-specific. This directly weakens one of the five named seed genes (*pou5f3/oct25/oct60*).
3. **Retain vs re-acquire is explicitly open** in the primary community's own synthesis  
(P 35534333):  
"either retained ... or subsequently reactivated").
4. **Function is pleiotropic, not NC-specific.** *id3/ID3* performs the same bHLH-sequestration and survival/pluripotency role in hESC, spermatogonia, erythroid progenitors and osteoblasts  
(P 35701409),  
(P 39342788),  
(P 42733085).  
A NC-specific GO term implies specificity the molecule does not have.
5. **Paralog/allele considerations.** Q91399 is the *X. laevis* **id3-a** homeolog (allotetraploid genome has *id3-a* and *id3-b*). Most classic functional work (Kee & Bronner 2005) predates explicit homeolog resolution and used *X. laevis* MO; Briggs 2018 is *X. tropicalis* (diploid, single *id3*). Curators should confirm orthology/homeology before transferring the *tropicalis* dataset claim to the *laevis* *id3-a* record.

## Knowledge Gaps

| Gap                                                                                                      | What was checked                                                                                                                                        | Why it matters                                                     | What would resolve it                                                                                                                                |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is expression truly <i>continuous</i> (no gap) for id3 across the blastula→NC trajectory in Briggs data? | Identified dataset (  ) and its stated scope; did <b>not</b> reanalyse | Core of seed's proposed test; distinguishes maintain vs re-acquire | Download GSE/EB-cloud Briggs <i>X. tropicalis</i> object; plot id3/myc/hes4/pou5/lin28 along the inferred NC trajectory; test co-expression per cell |
| Does id3-a (homeolog) specifically, vs id3-b, drive the NC phenotype?                                    | Noted homeolog ambiguity                                                                                                                                | GO annotation is per-gene (Q91399 = id3-a)                         | Homeolog-specific MO/CRISPR or allele-resolved scRNA in <i>X. laevis</i>                                                                             |
| Is "maintenance" mechanistically distinct from "proliferation/survival" for GO purposes?                 | Reviewed MF/BP evidence                                                                                                                                 | Determines whether a new term adds information                     | Direct test of whether id3 blocks differentiation independent of proliferation/survival in NC                                                        |
| Which bHLH partners does id3 sequester in NC?                                                            | MF is general (E-proteins/TCF3) in other tissues                                                                                                        | Pins down the MF annotation precisely in NC context                | Co-IP / proximity labeling of id3 in <i>Xenopus</i> NC                                                                                               |

## Discriminating Tests (most efficient)

- 1. Trajectory co-expression reanalysis (the seed's own test):** On the Briggs 2018 *X. tropicalis* object, order cells along the pluripotent→NC-border→premitratory-NC axis and quantify (a) fraction of cells co-expressing all five factors per stage, (b) presence/absence of an expression "gap" for each gene. *Interpretation caveat: a positive result supports continuity but still does not establish a maintenance function.*

2. **Perturbation-with-rescue (decisive for function):** Stage-specific id3-a depletion limited to the border (vs blastula) to test whether id3 is required to *maintain* an already-specified progenitor pool vs to *induce* it — distinguishes maintenance from induction.
3. **Decouple proliferation/survival from differentiation block:** Combine id3 loss with apoptosis/ cell-cycle rescue and ask whether NC identity is still lost — tests whether "maintenance" is more than keeping cells alive.
4. **Cross-species comparison:** Repeat continuity analysis in zebrafish/lamprey datasets to test the generality challenged by

**P** 39060477

## Curation Leads (require curator verification)

- **Candidate references to attach to the id3-a review:**

- **P** 15769946

— snippet to verify: *"Morpholino oligonucleotide-mediated depletion of Id3 results in the absence of neural crest precursors and a resultant loss of neural crest derivatives. This appears to be mediated by cell cycle inhibition followed by cell death of the neural crest progenitor pool, rather than a cell fate switch."* → supports BP: NC development / stem cell population maintenance / regulation of apoptotic process (IMP).

- **P** 39060477

— snippet: *"a lamprey pou5 orthologue is expressed in animal pole cells but is absent from neural crest"* → qualifier against universal continuity.

- **P** 35534333

— snippet: *"pluripotency is either retained in the neural crest from blastula stages or subsequently reactivated"* → flags the model as unresolved.

- **P** 29700227

— the dataset underpinning the proposed analysis.

- **Candidate GO actions:**

- *Do not* create/assign a standalone "neural crest progenitor maintenance" term as primary.

- *Assign/retain* MF = negative regulation of DNA-binding transcription factor activity (GO:0043433) and bHLH-sequestration binding (not bare "protein binding").
- *Assign* BP = neural crest development (GO:0014029) and stem cell population maintenance (GO:0019827), ideally with an annotation extension localizing to neural crest, supported by IMP from

**P** 15769946

- **Suggested curator questions:**
- Is the proposed term meant as MF, BP or CC? (It reads as BP but is being argued from expression, an MF/co-expression observation.)
- Does AIGR want tissue-specificity encoded as a new term or via annotation extensions on existing terms?
- Has id3-a vs id3-b homeolog specificity been confirmed for Q91399?
- **Suggested experiments:** the four discriminating tests above, prioritizing #2 (stage-specific perturbation) as the only one that directly tests *maintenance*.

## Provenance

- Reference identification

(

**P** 39060477

York 2024 Nat Ecol Evol;

**P** 29700227

Briggs 2018 Science) fetched programmatically via NCBI E-utilities (efetch) within `execute_code`; abstracts quoted above are verbatim from that fetch.

- Functional/mechanistic literature retrieved via `search_pubmed`; snippets quoted are verbatim from returned abstracts.
- Current GO annotation state fetched programmatically from UniProt REST (`rest.uniprot.org/uniprotkb/Q91399.json`, HTTP 200) within `execute_code`; the GO list and domain annotation (single HLH domain aa32–84; InterPro IPR026052 "DNA-binding protein inhibitor") are verbatim from that record. Decision table saved as `id3a_GO_decision_table.csv`.

- Briggs dataset accessibility confirmed via GEO brief record for **GSE113074** (HTTP fetch in `execute_code`): public, 136,966 *X. tropicalis* cells. **No scRNA reanalysis was executed** and no expression-continuity figure is presented, to avoid implying an analysis that was not run.
- Sequence analysis (Iteration 3): FASTA for Q91399 (Id3-a), Q02535 (human ID3) and P15172 (human MYOD1) fetched from UniProt; K/R-fraction window scan computed in `execute_code`. Result: Id3-a max 13-aa basic-region K/R fraction 0.15 vs MYOD1 0.54 — Id3 lacks a DNA-binding basic region. This is a genuine computed result, not a synthesized figure.

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